Answers and talk notes

Determine the Area of Triangles

What your student is learning

Tonight we work the triangle area formula BACKWARDS — given the area and one measure, we find the other.

Answers

Ask your student to explain before you look.

1. **C.** Double it. Double it: doubling the area undoes the half.
2. **A.** 4 cm. Double 12 to get 24, then 24 ÷ 6 = 4 cm.
3. **D.** 12 m. Double 30 to get 60, then 60 ÷ 5 = 12 m.
4. **A.** ½ × 9 × h = 18. Put the known numbers into the formula: ½ × 9 × h = 18.
5. **C.** 8 cm. Double 28 to get 56, then 56 ÷ 7 = 8 cm.

Talk about it

The area of a triangle is 12 and its base is 8. What do you do first to undo the half?

Home mission

Say, "A flag has an area of 45 square inches and a height of 9 inches." Ask your student to explain how to find the base. What do you do first? Then ask them to check the answer in the formula.

When they are stuck, ask

- What does the formula say, and which part is missing?
- What undoes taking half of something?
- Put your answer back in. Does the area come out right?

Try not to say

- Divide the area by the height and you are done.
- Forget the half, it barely changes anything.
- You do not need to check, the arithmetic was easy.